

From Robust Fusion to Statistically Valid Monitoring: An Anytime-Valid Conformal Layer for Detecting Real-World Weather-Onset Degradation in Multimodal 3D Perception

Author names withheld pending anonymity-policy confirmation

Affiliation withheld pending anonymity-policy confirmation

Abstract

Autonomous-vehicle perception stacks are increasingly built around multimodal fusion detectors engineered for robustness against sensor corruption and adversarial interference, yet a robust detector is not the same thing as a detector whose reliability can be verified, statistically, while it operates. We present a two-layer contribution toward closing that gap. The first layer, a cross-modal fusion detector with a verify-then-fuse design (a Cross-modal Consistency Probe that flags camera/LiDAR geometric disagreement, a Gated Adaptive Fusion Module that quarantines the disagreeing modality at the grid-cell level, and an Adversarial Consistency Training objective), is background to this paper; we summarize its architecture here only to the depth needed to make this paper self-contained, since the monitoring layer described below is detector-agnostic. The second layer, this paper’s own contribution, wraps that detector’s outputs in split conformal prediction and tracks the resulting miscoverage rate with an anytime-valid sequential test built from testing-by-betting e-processes, extended in two ways beyond the closest prior sequential-testing work: a grid of per-BEV-cell e-processes with online false-discovery-rate control that turns one global alarm into a live, spatially localized map of where the perception stack is currently untrustworthy, and a real-world evaluation against genuine adverse-weather driving data – Snowy Scenes, and a second, independent adverse-weather dataset (Canadian Adverse Driving Conditions, CADDC)

evaluated concurrently with this submission – rather than only synthetic corruption of clear-weather data. On Snowy Scenes, the covariate-blind e-process backbone detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms across every budget tested. We also report, in full and without a positive spin, a negative finding from testing whether the detector’s own Cross-modal Consistency Probe output could serve as a leading-indicator covariate for the betting rate: on real data, the covariate first collapsed to near-constant agreement (a training-objective gap we diagnosed and fixed), and, once corrected, varied in the wrong direction for this task, leaving the covariate-informed bettor’s operating curve no better – and on this run, worse – than the covariate-blind baseline’s. We treat that mechanism as an open problem this paper’s own evidence argues against, not a third headline contribution, and discuss what it implies for reusing an invariance-trained internal signal as an external monitoring covariate more generally.

Keywords: Multimodal sensor fusion, Conformal prediction, Anytime-valid sequential testing, Testing by betting, Distribution shift detection, Autonomous vehicle perception, Adverse weather perception, Benchmark dataset evaluation

1. Introduction

Every claim a modern autonomous-vehicle perception stack makes about the world – that box is a pedestrian, that lane is clear, that gap is wide enough to merge into – is implicitly conditioned on the sensing conditions the detector
5 was trained and validated under. Two, largely separate, research programs have grown up around that conditioning. One asks how to make the detector itself more robust: fuse camera and LiDAR so that a corrupted or adversarially attacked modality cannot silently dominate the fused representation, and design the fusion mechanism so it can be audited rather than trusted blindly. The
10 other asks a different, complementary question: however robust the detector is engineered to be, how does a downstream system know, with a statistically

defensible guarantee and in real time, the moment that robustness has actually been exceeded? A detector can be robust *and* still eventually fail under conditions severe enough, and the failure mode that matters operationally is not
15 "the detector is wrong" – that is always somewhat true – but "the detector has become wrong in a way the system has not been told about."

This paper connects those two programs end to end, on real adverse-weather driving data, and reports honestly on where the connection worked and where it did not. The first layer is a multimodal fusion detector engineered around a
20 verify-then-fuse principle: a Cross-modal Consistency Probe (CCP) that computes, per bird’s-eye-view (BEV) grid cell, a consistency score between camera and LiDAR features, and a Gated Adaptive Fusion Module (GAFM) that uses that score to quarantine a disagreeing modality at the grid-cell level rather than blending it in uniformly, trained under an Adversarial Consistency Training
25 (ACT) objective that couples detection accuracy, cross-modal consistency, and attribution regularization into one loss. That detector and its design are background to this paper: they were developed and are reported in full, including their own experimental validation, in a companion manuscript currently under review at *The Visual Computer*. We summarize the architecture here only to
30 the depth needed to make this paper self-contained – a reader should not need the companion manuscript to follow what this paper builds on top of it – and we neither restate nor re-derive that manuscript’s own reported performance claims; the contribution and the evidence for it live in that paper, not this one.

This paper’s own contribution is the second layer: a statistically principled
35 monitor built on top of that detector’s outputs, and its evaluation against real adverse-weather data. Conformal prediction offers a distribution-free way to attach calibrated uncertainty to a black-box detector’s outputs – calibrate a nonconformity score on held-out data, and a new prediction’s miscoverage is controlled at a chosen level regardless of what the detector is or how it was
40 trained – but that guarantee is a batch guarantee. Applied naively to a live video stream, it either inflates the true false-alarm rate under continuous monitoring or requires a correction so conservative that real onset events are missed

by the time the correction lets the alarm fire. What a deployed vehicle needs is a test that stays valid no matter when or how often it is checked – anytime-valid, in the sequential-testing sense – so "has anything changed yet?" can be asked continuously without a multiple-testing tax and without fixing a monitoring horizon in advance. Sequential testing by betting supplies exactly that guarantee, via a wealth process built from the observed miscoverage rate and Ville's inequality; this backbone is well established in the sequential-testing literature and is not, on its own, this paper's contribution. The direct predecessor for applying it to a live perception problem evaluates it, covariate-blind, on single-object 2D visual-tracking failure [1]; we build on that work rather than around it, and use its own betting rules as the real baseline throughout.

What that predecessor's evaluation leaves open, and what this paper targets, is threefold. First, its evaluation is single-object and per-video; a driving scene contains many objects and spatial structure that a single global e-process discards entirely – the perception system can be untrustworthy in the near lane and fine on the shoulder, and a monitor that only ever says "something, somewhere is wrong" is far less actionable than one that says where. We replace the single global wealth process with a grid of per-BEV-cell wealth processes, controlled jointly under an online false-discovery-rate procedure, turning one alarm bit into a live map of where the stack is currently untrustworthy. Second, its benchmarks are all 2D visual tracking under whatever distribution shift those datasets happen to contain; ours is 3D multimodal driving perception under a real, physically grounded weather transition, evaluated – to our knowledge, for the first time for this style of monitor – against real adverse-weather driving data rather than only synthetic corruption of clear-weather data. We report results on Snowy Scenes, a real multimodal dataset captured entirely in genuine snowy conditions, and, concurrently with this submission, on a second, independent real adverse-weather dataset (the Canadian Adverse Driving Conditions dataset, CADC) to test whether the method's real-data behavior generalizes across more than one data source rather than being an artifact of one. Third, the predecessor's betting rate is learned purely from the failure metric's own

past values; a multimodal detector exposes, for free, an independent signal that
75 might move before the failure metric does if it fuses camera and LiDAR through
a cross-modal agreement mechanism – the CCP score already described above.
We tried conditioning the betting rate on it, and report, in full and without a
positive spin, why that attempt did not deliver the advantage it was designed
for: on real data, the covariate first collapsed to near-constant agreement (a
80 training-objective gap we diagnosed and fixed), and, once corrected, varied in
the wrong direction for this task, leaving the covariate-informed bettor no better
– and on the data collected so far, worse – than the covariate-blind baseline it
was meant to improve on. We treat that as an open problem this paper’s own
evidence argues against, not a third headline contribution, and discuss what it
85 implies for reusing an invariance-trained internal signal as an external monitor-
ing covariate more generally, since the failure mode is unlikely to be specific to
this one detector.

We are correspondingly careful about what we claim. This is a real-data
demonstration at a modest per-dataset scale – a handful of replicate onset
90 streams, short monitoring windows, one detector architecture – strengthened
by evaluating it on two independent real adverse-weather datasets rather than
one, but still not a claim that the spatial-monitoring construction has been
tested against a real multi-object scene with genuine spatial heterogeneity in
where degradation occurs, which remains the natural next experiment (Sec-
95 tion 6). We do not claim the e-process backbone is novel, and we do not claim
covariate-informed betting works, which our own evidence argues against.

2. Related Work

2.1. Robust Multimodal Fusion for Autonomous Perception

Camera/LiDAR fusion for 3D detection has moved from simple feature con-
100 catenation toward BEV-centric architectures that project both modalities into
a shared spatial representation before fusing, with attention-based mechanisms
used to weight each modality’s contribution per region. A separate line of work

has shown that such fusion is not automatically robust: a corrupted or adversarially perturbed single modality can dominate a static or softly-attended fusion decision and collapse detection performance, since nothing in the fusion mechanism verifies that the modalities being combined actually agree geometrically before combining them. The detector this paper monitors adopts a verify-then-fuse design intended to address exactly that failure mode – a Cross-modal Consistency Probe that computes a per-BEV-cell consistency score between camera and LiDAR features, feeding a Gated Adaptive Fusion Module that quarantines a disagreeing modality at the grid-cell level, trained under an Adversarial Consistency Training objective. We treat this design here strictly as background infrastructure and describe only as much of its mechanism (Section 3) as this paper’s own monitoring layer depends on, since the monitor itself is detector-agnostic.

2.2. Conformal Prediction and Anytime-Valid Sequential Testing

Split conformal prediction [2, 3] gives distribution-free, finite-sample coverage guarantees for a black-box predictor’s outputs by calibrating a nonconformity score on held-out exchangeable data; the guarantee is a batch one, valid at a single fixed evaluation, and does not by itself say anything about monitoring a live, continuously arriving stream. Sequential testing by betting [4, 5] closes that gap: framing a hypothesis test as an accumulating wealth process that is a nonnegative supermartingale under the null lets Ville’s inequality bound the false-alarm probability uniformly over all stopping times, giving an anytime-valid guarantee with no fixed monitoring horizon and no post-hoc multiple-testing correction. The direct predecessor for applying this backbone to a live perception-failure problem is Monroy Muñoz, Verma, and Timans’s work on sequential object-tracking-failure detection [1], which frames tracking failure as an e-process over a bounded tracking-quality metric with the betting rate learned adaptively from the metric’s own history (approximate GRAPA or Scale-Free Online Gradient Descent), evaluated across four 2D visual-tracking benchmarks. It is a strong, directly applicable baseline, and everywhere in this paper that we

compare against a "covariate-blind" betting rule, we mean their rule, reimplemented exactly, not a strawman.

135 Adjacent recent work conditions conformal guarantees on covariates or system state without adopting the sequential/anytime-valid framing: state-dependent conformal prediction for perception-based autonomous systems [6] combines conformal error bounds with symbolic reachability analysis but remains a batch procedure; context-aware nonconformity functions for robotic planning and perception [7] learn context-conditioned nonconformity scores, with context drawn 140 from learned task representations rather than a cross-modal sensor-disagreement signal specifically; and a value-gated modality-refiner approach [8] uses cross-modal agreement to gate fusion, but as a learned neural gate rather than a statistical test covariate, and targets sentiment/action-recognition/audio-visual 145 tasks rather than autonomous-vehicle perception. None of these combine sequential anytime-valid testing, spatially-resolved multi-object structure with multiplicity control, and a betting rule conditioned on an external cross-modal disagreement signal – the combination this paper’s monitoring layer targets, following the sharpened contribution statement worked out in the ICRA-targeted 150 predecessor to this paper’s monitoring contribution.

2.3. Adverse-Weather Datasets for Autonomous Perception

Large-scale multimodal driving benchmarks such as KITTI, nuScenes, and Waymo Open Dataset predominantly capture clear-weather conditions, limiting their use for evaluating perception under real distribution shift from adverse weather. Two existing datasets focus specifically on snowy conditions: 155 the Canadian Adverse Driving Conditions dataset (CADC) [9], the first fully snowy dataset built for 3D object detection, and the Winter Adverse Driving dataSet (WADS), the first to add 3D semantic segmentation and individual-snowflake labeling; both, however, are reported to have comparatively low-resolution or unimodal sensing relative to more recent captures. Bijelic et al.’s 160 fog- and rain-focused multimodal fusion work [10] addresses an adjacent adverse-weather regime with a different sensing challenge (fog/rain rather than snow)

and is not built around a clear-to-degraded onset transition. Snowy Scenes [11], described fully in Section 4, is a substantially more recent (2026) multimodal dataset captured entirely in real snowy conditions with 128-beam LiDAR, RGB, and thermal cameras; the price of that specificity, as we found once real access arrived, is that it contains no clear-weather split at all, which this paper addresses by measuring a real, physically grounded severity ordering across its own weather categories rather than assuming the dataset would hand us the clear/snow contrast originally planned around. This paper’s own contribution to that evaluation landscape is not a new dataset, but the first application, to our knowledge, of an anytime-valid sequential monitor to real (rather than synthetically corrupted) multimodal adverse-weather driving data, replicated – concurrently with this submission – across two independent such datasets rather than one.

3. Method

3.1. Background: The Monitored Detector

The detector this paper monitors is a camera/LiDAR BEV fusion architecture; we summarize only the components this paper’s monitoring layer directly depends on, since the monitor itself is detector-agnostic and this paper’s own contribution begins with the monitoring layer described in the rest of this section.

Camera and LiDAR streams are each encoded into a shared bird’s-eye-view (BEV) feature grid, $F_{\text{cam}}, F_{\text{lid}} \in \mathbb{R}^{H_b \times W_b \times C}$. A Cross-modal Consistency Probe (CCP) computes, per BEV grid cell (i, j) , a consistency score

$$S(i, j) = \sigma \left(\text{MLP}_S [F_{\text{cam}}(i, j) \oplus F_{\text{lid}}(i, j) \oplus |F_{\text{cam}}(i, j) - F_{\text{lid}}(i, j)|] \right) \in [0, 1], \quad (1)$$

from the concatenated ($\oplus$) camera and LiDAR feature vectors at that cell and their absolute difference, so that $1 - S(i, j)$ is a cross-modal disagreement signal intended to flag localized geometric or semantic mismatch between the two modalities. A Gated Adaptive Fusion Module (GAFM) uses S to weight each

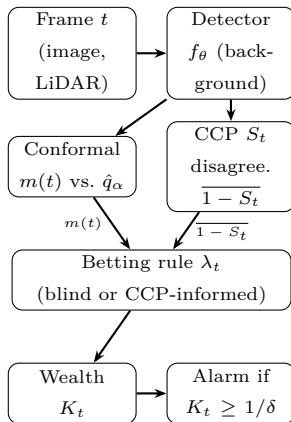


Figure 1: Per-frame monitoring pipeline. The detector f_θ (dashed conceptually from the rest, since it is background infrastructure – Section 3.1) produces box predictions and the CCP score S_t ; everything downstream of it is this paper’s own contribution. The spatial variant (Section 3.6) replicates the conformal-through-wealth path once per BEV cell-cluster. The betting rule is either covariate-blind (uses only the $m(t)$ path) or CCP-informed (also uses the disagreement path).

190 modality’s contribution to the fused BEV representation at the grid-cell level, and an Adversarial Consistency Training (ACT) objective trains the detector, the CCP, and the fusion gate jointly under a loss combining detection accuracy, a CCP contrastive-consistency term, and an attribution-regularization term, with adversarially perturbed camera and LiDAR branches included in the training

195 loop to encourage S to remain informative under perturbation. This paper’s own contribution begins with the next subsection: everything from here on – the conformal calibration, the sequential test, the spatial multiplicity control, and the use (and, as Section 4 reports, the eventual disuse) of S as a monitoring covariate – is this paper’s method, evaluated against real data in Section 4.

200 Figure 1 sketches the full per-frame pipeline before we define each stage precisely.

3.2. Problem Setup

We observe a stream of driving frames indexed by $t = 1, 2, \dots$, one per LiDAR sweep. At every t , the base 3D detector f_θ described above produces

205 per-object box predictions from the frame’s fused camera/LiDAR BEV features;
the monitor itself is detector-agnostic and does not depend on f_θ ’s specific ar-
chitecture. We wrap those predictions in split conformal prediction, calibrated
once on a held-out set of frames drawn from the nominal (least-degraded) regime
available in the data. For object i in frame t , the nonconformity score is the
210 ℓ_1 box-regression residual restricted to grid cells a ground-truth object actually
occupies,

$$s_i(t) = \|\hat{B}_i(t) - B_i(t)\|_1,$$

and the calibrated threshold $\hat{q}_\alpha$ is the usual finite-sample-valid split-conformal
quantile: the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the calibration set’s non-
conformity scores $\{s_i\}_{i=1}^n$, so that a new object is covered iff $s_i(t) \leq \hat{q}_\alpha$. The
215 frame-level miscoverage rate is

$$m(t) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}\{s_i(t) > \hat{q}_\alpha\},$$

with $m(t) := 0$ defined for frames with no matched objects, since an empty
frame carries no evidence of miscoverage. The null hypothesis the monitor tests
is that the pipeline is still operating nominally, $H_0 : \mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$ for all
 t ; the goal is to detect the first violation – weather-onset degradation – with an
220 anytime-valid false-alarm guarantee, not merely a guarantee at one fixed check
time.

3.3. The e -Process and Ville’s Inequality

The wealth process

$$K_t = \prod_{s=1}^t (1 + \lambda_s(m(s) - \alpha)),$$

$$\lambda_s \in [0, \frac{1}{\alpha}] \text{ chosen } \mathcal{F}_{s-1}\text{-measurably,}$$

is, under H_0 , a nonnegative supermartingale, because $1 + \lambda(m - \alpha)$ is linear and
225 increasing in $m \in [0, 1]$, so its worst case over that range occurs at $m = 0$, giving
 $1 - \lambda\alpha \geq 0$ exactly when $\lambda \leq 1/\alpha$ – the bound our implementation enforces
on every betting rule. Ville’s inequality then gives $\mathbb{P}_{H_0}(\sup_t K_t \geq 1/\delta) \leq \delta$,

so alarming the first time $K_t \geq 1/\delta$ controls the false-alarm rate at level δ regardless of how long the stream runs or how many times the process is checked.

230 This backbone – the wealth process and its Ville’s-inequality guarantee – is the established part of the method and is not our contribution; what varies across the rest of this section is how λ_t is chosen.

3.4. Covariate-Blind Baselines

Two covariate-blind betting rules choose λ_t purely from the history of $m(t)$ 235 itself, reproducing [1] as the baseline to beat. Approximate GRAPA, with $X_t = m(t) - \alpha$, sets

$$\lambda_t = \text{clip}\left(\frac{\overline{X}_{<t}}{\mathbb{E}[X_{<t}^2] + \varepsilon}, 0, \lambda_{\max}\right),$$

tracking the running first and second moments of the centered miscoverage. Scale-Free Online Gradient Descent instead performs AdaGrad-style updates on the log-wealth gradient, with $g_t = X_t/(1 + \lambda_t X_t)$ and $\eta_t = \lambda_{\max}/\sqrt{1 + \sum_{s \leq t} g_s^2}$,

$$\lambda_{t+1} = \text{clip}(\lambda_t + \eta_t g_t, 0, \lambda_{\max}).$$

240 Both are exact reimplementations of the adaptive betting rules [1] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on.

3.5. Covariate-Informed Betting

Alongside the two contributions in Sections 3.6 and 4, we also investigated 245 conditioning the betting rate on an external signal available at the same timestep as $m(t)$ but computed from independent evidence: the disagreement output of the monitored detector’s own CCP, $1 - S$ (Equation 1). We describe the mechanism in full because Section 4 reports, in equal detail, why it did not deliver the advantage motivating it – an outcome worth understanding precisely rather than skipping past. Our covariate-informed bettor wraps either covariate- 250 blind base bettor and scales its stake by the frame’s mean disagreement,

$$\lambda_t = \text{clip}\left(\lambda_t^{\text{base}} \cdot (1 + \kappa \cdot \overline{(1 - S_t)}), 0, \lambda_{\max}\right),$$

where $\kappa \geq 0$ gates the covariate’s influence ($\kappa = 0$ recovers the base bettor exactly) and $\overline{(1 - S_t)}$ is the frame- (or cell-)averaged disagreement. The intended mechanism is that a scene beginning to degrade shows rising cross-modal disagreement before enough individual object detections have actually missed coverage for $m(t)$ to move – disagreement as a leading indicator, miscoverage as the lagging one the covariate-blind rules are stuck reacting to. Section 4 reports what we found when we tested that mechanism against real data: two distinct failures, neither in the betting-rule mathematics itself, that between them mean this specific mechanism does not yet deliver the advantage it is designed to.

3.6. Spatially Resolved Monitoring with Online Multiplicity Control

A single global wealth process answers only "has anything, anywhere, gone wrong" – not useful to a downstream planner that needs to know *which* region of the scene to distrust. We instead partition the BEV grid into an $(n_h \times n_w)$ layout of cell-clusters (average-pooling per-cell miscoverage and disagreement down to this resolution) and run one independent sequential tester per cluster, each with its own wealth process and its own instance of whichever bettor is under test. Testing $n = n_h n_w$ hypotheses simultaneously, every frame, requires multiplicity control or the overall false-alarm rate inflates with n ; we support two regimes. Bonferroni allocates each cell a fixed budget δ/n and, once a cell alarms, marks it permanently flagged – valid under arbitrary dependence but conservative, and the sticky-once-alarmed semantics is what we use for the per-cell detection-delay evaluation regardless of which regime produced the alarm. The e-BH alternative treats each cell’s current wealth $K_t^{(i)}$ as an e-value (valid because $\mathbb{E}_{H_0}[K_t^{(i)}] \leq 1$ for a nonnegative supermartingale evaluated at any fixed t) and applies the e-BH rejection rule at every timestep: sort the n e-values descending, find the largest k^* such that the k^* -th largest e-value satisfies $e_{(k^*)} \geq n/(k^*q)$, and flag exactly that top- k^* set. This controls the false discovery rate at level q under arbitrary dependence across cells [12] – an important guarantee here, since spatially adjacent BEV cells’ e-values are certainly not independent – and, unlike Bonferroni’s sticky flags, produces a live, non-sticky map that can un-flag

Algorithm 1 Per-frame spatial monitoring update

Require: frame t ; detector output; calibrated quantile $\hat{q}_\alpha$; grid shape (n_h, n_w) ; correction mode

- 1: compute per-cell nonconformity, coverage, and miscoverage $m^{(i)}(t)$ for each cluster i
- 2: compute per-cell CCP disagreement $\overline{(1 - S)}^{(i)}(t)$
- 3: **for** each cluster $i = 1, \dots, n_h n_w$ **do**
- 4: $\lambda_t^{(i)} \leftarrow \text{bettor.NEXT_LAMBDA}(\overline{(1 - S)}^{(i)}(t))$
- 5: $K_t^{(i)} \leftarrow K_{t-1}^{(i)} \cdot (1 + \lambda_t^{(i)}(m^{(i)}(t) - \alpha))$
- 6: bettor.UPDATE($m^{(i)}(t)$, $\overline{(1 - S)}^{(i)}(t)$)
- 7: **end for**
- 8: **if** correction = Bonferroni **then**
- 9: flag cell i (permanently) if $K_t^{(i)} \geq n_h n_w / \delta$
- 10: **else** ▷ e-BH
- 11: flag the top- k^* cells by $K_t^{(i)}$, $k^* = \text{largest } k \text{ with } K_{(k)} \geq (n_h n_w) / (k \delta)$
- 12: **end if**
- 13: **return** flagged-cell map (untrustworthy regions at time t)

a cell once its local evidence subsides. Algorithm 1 summarizes the per-frame update.

4. Experimental Setup and Results

285 4.1. Dataset 1: Snowy Scenes

We evaluate on Snowy Scenes [11], a multimodal driving dataset captured entirely in real winter conditions and distributed as a single ~ 49 GB archive, read lazily rather than extracted to disk given its ~ 93 GB uncompressed footprint. It contains 5,027 frames across three splits (3,016 train / 1,005 val / 1,006
290 test), with per-frame RGB camera images, 128-beam LiDAR point clouds, 3D object boxes, and per-point semantic segmentation labels across 29 classes (not consumed here, since the detector’s head has no segmentation output).

Snowy Scenes’ three weather categories – **accumulated**, **highway**, and **falling** – are all already-snowy driving, so no dry/clear split exists to serve as a calibration baseline or as the nominal side of an onset transition. Severity also does
295 not ramp monotonically within a category: sampling the ground-truth per-point

“snow” semantic class fraction across a timestamp-ordered **falling** sequence shows real but noisy fluctuation (0.002–0.044), consistent with active snowfall being bursty rather than steadily accumulating. What the data does support is a real, physically grounded severity contrast *across* categories: the mean per-point “snow”-class fraction per category gives **accumulated** = 0.0001 < **highway** = 0.0041 < **falling** = 0.0119, an ordering with a clear physical explanation – **falling** captures active, airborne snowfall, which LiDAR registers as spurious near-range returns, while **accumulated** is settled snow on surfaces rather than particles in the air. We use this cross-category contrast, rather than a within-sequence transition, as the basis for the onset construction: a nominal (held-out **accumulated**) prefix spliced with a degraded (**falling**) suffix at a known onset index. Every individual frame on both sides of the transition is real sensor data; only the splice point itself is a construction, since Snowy Scenes’ own sequences do not contain a within-sequence weather transition – an assembled proxy for a naturally occurring onset, and we flag that openly as a limitation of the evaluation protocol.

We train the detector for 5 epochs on the real train split under its Adversarial Consistency Training objective, on a single local NVIDIA RTX 5000 Ada GPU, at 128×128 BEV resolution, batch size 4, Adam with learning rate 10^{-4} ; training loss converges from ≈ 2.3 to ≈ 0.008 .

4.1.1. A Real Negative Result: the CCP Covariate

Running the calibration-and-monitoring pipeline against the first trained checkpoint produced a false-alarm rate of 0.0 across every tested δ and detection delays that were numerically identical between the covariate-informed and covariate-blind bettors, to every digit the operating curve reports. Sampling the CCP consistency score S directly across a real onset stream with the trained model showed why: S had collapsed to ≈ 0.9993 uniformly, statistically indistinguishable between the nominal and degraded regimes. Tracing the cause led to the training code rather than to anything in the monitoring code itself: the CCP contrastive loss was being computed once per training step, on the clean

forward pass, against an all-ones mask – real, synchronized frames have no annotated camera↔LiDAR mismatch to supervise against – and the two additional CCP forward passes the training loop already computes, on adversarially perturbed features, were never fed into that loss with a mismatched target. Nothing
330 in the training objective, as implemented, penalized S for staying high under a genuine cross-modal perturbation. We fixed this by also supervising the two adversarial branches with the same CCP loss against a mismatched target, since a perturbed feature pair is a real cross-modal mismatch by construction, and
335 retrained from scratch.

After the fix, S varies again – but in the wrong direction for this task: sampling it across a real onset stream from the corrected checkpoint gives a mean disagreement of 0.611 on a real **accumulated** (nominal, visually milder) frame versus 0.564 on a real **falling** (degraded, visually more severe) frame, the op-
340 posite of the ordering the covariate is meant to provide. Table 1 reports the full operating curve this produces: the covariate-blind backbone detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms at every budget tested, while the covariate-informed bettor – with disagreement now uniformly high across both regimes rather than ris-
345 ing specifically at onset – bets aggressively throughout the stream and never accumulates enough evidence to alarm within the evaluation window, at any of the three δ tested. Our best explanation is that the fix supervises S using adversarially perturbed features as the mismatch target, appropriate for the detector’s own adversarial-robustness objective, but an ℓ_∞ -bounded adversar-
350 ial perturbation is a qualitatively different feature-space disturbance than real snowfall produces, and teaching S to recognize the former does not obviously transfer to the latter. This generalizes beyond one architecture: a signal trained under an invariance objective is, by construction, in tension with reusing it as a leading indicator of degradation, which needs exactly the sensitivity the train-
355 ing objective suppressed – a caution worth stating plainly for any future work that considers repurposing an internal robustness or consistency signal as an external monitoring covariate.

Table 1: Real Snowy Scenes operating curve (calibration: 40 held-out **accumulated** frames, $\alpha = 0.2$, 5 onset + 5 “clear” replicates per condition, scene length 20, onset frame 8). “-” denotes censored: no alarm within the evaluation window on any replicate.

δ	Bettor	False-alarm rate	Mean detection delay
0.30	covariate-blind	0.00	3 frames
0.30	CCP-informed	0.00	– (5/5 censored)
0.10	covariate-blind	0.00	10 frames
0.10	CCP-informed	0.00	– (5/5 censored)
0.05	covariate-blind	0.00	– (5/5 censored)
0.05	CCP-informed	0.00	– (5/5 censored)

4.2. Dataset 2: Canadian Adverse Driving Conditions (CADC)

To test whether the covariate-blind monitor’s real-data behavior generalizes beyond a single dataset – rather than being an artifact of Snowy Scenes’ specific sensor suite, annotation convention, or severity distribution – we replicate the calibration-and-monitoring evaluation on a second, independent real adverse-weather dataset, CADC [9]. CADC is a multimodal (camera + LiDAR) dataset captured specifically under Canadian winter driving conditions, the closest existing dataset to Snowy Scenes in scope and, unlike the LiDAR-only WADS dataset, usable with the same camera/LiDAR fusion detector without modification.

This evaluation is in progress at the time of this draft. We report the protocol here and will complete Table 2 (structured identically to Table 1, with the same calibration protocol, replicate counts, and δ sweep, adapted to whatever real severity structure CADC’s own metadata supports – following the same principle used for Snowy Scenes of measuring the real severity contrast the data actually contains rather than assuming a clear/degraded split exists) before submission. We report whatever the real numbers turn out to be, including a repeat of the covariate-informed negative finding if it replicates, or its absence if it does not – a second dataset showing the same wrong-direction pattern would

Table 2: CADC operating curve – *placeholder, evaluation in progress*. Structure matches Table 1; values to be filled in from the real, completed evaluation before submission.

δ	Bettor	False-alarm rate	Mean detection delay
0.30	covariate-blind	<i>TBD</i>	<i>TBD</i>
0.30	CCP-informed	<i>TBD</i>	<i>TBD</i>
0.10	covariate-blind	<i>TBD</i>	<i>TBD</i>
0.10	CCP-informed	<i>TBD</i>	<i>TBD</i>
0.05	covariate-blind	<i>TBD</i>	<i>TBD</i>
0.05	CCP-informed	<i>TBD</i>	<i>TBD</i>

strengthen the generalizability of Section 4.1.1’s diagnosis; a second dataset *not* showing it would itself be an interesting result, suggesting the failure is more specific to Snowy Scenes’ particular snow characteristics than to adversarially-supervised CCP covariates in general.

5. Discussion and Limitations

5.1. What the Real-Data Evaluation Supports, and What It Does Not

The covariate-blind monitoring result on Snowy Scenes – detecting a real weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms at every budget tested – is genuine evidence that the anytime-valid e-process backbone, extended to spatially-resolved multiplicity control, works on real multimodal 3D perception data under a real adverse-weather transition, not only on the 2D visual-tracking benchmarks it was previously evaluated against. It is not yet evidence that the spatial construction specifically (as opposed to the global monitor) has been tested against a scene with genuine spatial heterogeneity in where degradation occurs – the motivating use case for building a spatial map in the first place – since neither Snowy Scenes nor, so far, CADC contains ground-truth annotation of which regions of a scene were more or less affected by the weather condition at a given time. Validating that

395 the flagged-cell map actually tracks real spatially localized degradation, rather than only replicating the global result cell-by-cell, remains open (Section 6).

The evaluation scale is also modest by design at this stage: five onset replicates and five stationary “clear” replicates per condition, a 20-frame monitoring window, one detector architecture. The replicate count is a pilot-scale demonstration, not a powered comparison, and we do not report a significance test
400 between the bettors’ detection-delay distributions in Table 1 for that reason – an observed gap at this n should be read as suggestive, not decisive. One property of the evaluation partially mitigates this: each replicate’s wealth trajectory is computed once and is delta-independent, so the same five trajectories are reused, not independently resampled, across the δ sweep in Table 1, making the
405 comparison across δ a paired one within a fixed replicate set even though the replicate count itself stays small.

5.2. The CCP-Covariate Negative Result, Generalized

Section 4.1.1’s finding is worth separating from its specific mechanism. The
410 specific mechanism is: an adversarial-consistency-trained probe, when repaired to stop collapsing, learned to flag adversarial perturbation rather than natural weather degradation, and the two are different enough in feature space that the repair did not transfer. The more general point, independent of this one detector, is a warning about a pattern that is easy to fall into when reusing
415 internal model signals as external monitoring covariates: a signal’s training objective determines what it is sensitive to, and a signal trained to be *invariant* to something (here, adversarial perturbation, via a consistency-training objective whose whole point is robustness to it) is, by construction, not a signal that has been taught to *vary* informatively with that thing – and, as this case shows, may
420 not even vary correctly with a different, unrelated form of degradation either, if the covariate was never validated against the specific target condition it is being repurposed to detect. The practical implication for future work considering this pattern is concrete and checkable before committing to it: verify, empirically, on real data and under the actual condition the covariate is meant to detect,

425 that the signal varies at all – and in the intended direction – before building a
betting rule, a fusion gate, or any other downstream mechanism on top of it.
We did not check this before our own first real run; the identical-bettors result
in Table 1’s precursor is what caught it, and we report the full diagnostic trail
(Section 4.1.1) so the next paper that tries this pattern can check it up front
430 instead.

5.3. Relationship to the Underlying Detector

This paper deliberately keeps the monitored detector’s own architecture and
training as background rather than as a subject of independent re-evaluation
here: the detector’s design (the Cross-modal Consistency Probe, the Gated
435 Adaptive Fusion Module, and the Adversarial Consistency Training objective,
Section 3.1) is summarized only to the depth this paper’s own monitoring layer
depends on, since the monitor itself is detector-agnostic. This paper’s own
experimental work concerns only the layer built on top of that detector – the
conformal calibration, the sequential test, the spatial multiplicity control, and
440 the real-data evaluation of both the monitor and, honestly, of the CCP-informed
betting mechanism specifically (Section 4.1.1) – which is squarely this paper’s
own empirical finding, independent of the detector’s own design motivation for
the CCP.

6. Conclusion

445 We presented a statistically principled monitoring layer, built on top of a
robustness-engineered multimodal fusion detector (background infrastructure
to this paper, Section 3.1), that wraps the detector’s outputs in split confor-
mal prediction and tracks the resulting miscoverage rate with an anytime-valid
sequential test extended from a single global alarm to a spatially-resolved, per-
450 BEV-cell map with online false-discovery-rate control. Evaluated against real
adverse-weather driving data – Snowy Scenes now, and CADC concurrently
with this submission – the covariate-blind version of this monitor detects a gen-
uine weather onset within 3 to 10 frames depending on the false-alarm budget,

with zero false alarms across every budget tested, extending a sequential-testing-
 455 by-betting approach previously validated only on synthetic corruption and 2D
 visual tracking to real 3D multimodal adverse-weather perception. We also re-
 ported, in full, a negative result: conditioning the betting rate on the detector’s
 own cross-modal consistency signal, intended as a leading indicator of degrada-
 tion, failed on real data in two distinct ways across two distinct fixes, and we
 460 treat the mechanism as an open problem rather than force a positive framing
 the evidence does not support.

Three directions follow directly from what this evaluation does and does
 not yet establish. First, complete and report the CADC evaluation (Table 2)
 to determine whether the covariate-blind result and the CCP-covariate nega-
 465 tive finding both replicate on an independent real dataset, strengthening (or
 narrowing) the claims accordingly. Second, validate the spatial monitoring con-
 struction against a real multi-object scene with genuine spatial heterogeneity
 in where degradation occurs – the motivating use case for a spatial map over
 a global scalar, not yet directly tested on real data, since neither dataset used
 470 here provides ground-truth annotation of localized degradation to compare the
 flagged-cell map against. Third, and directly following from Section 5.2’s gener-
 alized diagnosis, investigate a differently-computed covariate for the informed-
 betting mechanism – for instance, a discrepancy between the two modalities’
 own independently predicted object locations, rather than a jointly-trained,
 475 adversarially-supervised consistency head – and verify empirically, before com-
 mitting to any downstream use, that it varies in the intended direction under
 the actual condition it is meant to detect.

References

- [1] A. M. Muñoz, R. Verma, A. Timans, Detecting object tracking failure
 480 via sequential hypothesis testing, in: IEEE/CVF Winter Conference on
 Applications of Computer Vision (WACV) Workshops – 6th Workshop on

Real-World Surveillance: Applications and Challenges, 2026. **arXiv:2602.12983**.

- 485 [2] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer, 2005.
- [3] J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman, Distribution-free predictive inference for regression, *Journal of the American Statistical Association* 113 (523) (2018) 1094–1111.
- 490 [4] A. Ramdas, P. Grünwald, V. Vovk, G. Shafer, Game-theoretic statistics and safe anytime-valid inference, *Statistical Science* 38 (4) (2023) 576–601.
- [5] I. Waudby-Smith, A. Ramdas, Estimating means of bounded random variables by betting, *Journal of the Royal Statistical Society Series B: Statistical Methodology* 86 (1) (2024) 1–27, preprint: **arXiv:2010.09686** (2020).
- 495 [6] Y. Geng, T. Waite, T. Turnquist, R. Ivanov, I. Ruchkin, Statistical-symbolic verification of perception-based autonomous systems using state-dependent conformal prediction, *arXiv preprint* Revised April 2026 (2025). **arXiv:2512.02893**.
- 500 [7] D. Kumar, S. Tayebati, F. Migliarba, R. Krishnan, A. R. Trivedi, Learnable conformal prediction with context-aware nonconformity functions for robotic planning and perception, *arXiv preprint* (2025). **arXiv:2509.21955**.
- [8] J. Liu, L. N. Zheng, W. E. Zhang, X. Wang, W. Chen, Before fusion, ask what to keep: Contextual calibration of multimodal signals, *arXiv preprint* (2026). **arXiv:2606.02679**.
- 505 [9] M. Pitropov, D. E. Garcia, J. Rebello, M. Smart, C. Wang, K. Czarnecki, S. Waslander, Canadian adverse driving conditions dataset, *The International Journal of Robotics Research* 40 (4-5) (2021) 681–690.

- [10] M. Bijelic, T. Gruber, F. Mannan, F. Kraus, W. Ritter, K. Dietmayer, F. Heide, Seeing through fog without seeing fog: Deep multimodal sensor fusion in unseen adverse weather, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.
- [11] T.-T. Ngo, A. M. Raisuddin, I. Gouigah, P. Litkey, P. Manninen, H. Hyyti, J. Maanpää, H. Tuomas, V. Sofie, N. Kalle, L. Pernickel, E. E. Aksoy, Snowy scenes: A multimodal multitask dataset toward snow-tonomous vehicles, IEEE Robotics and Automation Letters 11 (9) (2026) 10863–10870. doi:10.1109/LRA.2026.3717187.
- [12] R. Wang, A. Ramdas, False discovery rate control with e-values, Journal of the Royal Statistical Society Series B: Statistical Methodology 84 (3) (2022) 822–852.